Today's Agenda

Build simple OLS regressions for your research project report.

Justin Leinaweaver (Fall 2021)

Assignment 3: The Research Report

1. Introduction
   - What is the question, why is it important and what is your argument?
2. **Bivariate Analyses**
   - **What is the answer to the question using only bivariate visualizations and simple OLS regressions?**
3. Multivariate Analyses
   - What is the answer to the question using multivariate visualizations and multiple OLS regressions?
4. Conclusion
   - What did you find and how confident should we be in your results?

Research Project: Simple OLS Regressions

Fit, evaluate and make predictions for the following regressions:

1. Regress `fh_total` on `milper_prop`
2. Regress `pr` on `milper_prop`
3. Regress `cl` on `milper_prop`

Then, **rebuild each model** with a one period time lag on `milper_prop` (`milper_prop_lag1`)

Adding a Time Lag to a Variable

```r
## This code is already in the project Rmd file
d2 |>
  mutate(
    milper_prop_lag1 = lag(milper_prop, n = 1)
  )
```

Country	year	fh_total	milper_prop	milper_prop_lag1
Afghanistan	1973	13	0.6180386	NA
Afghanistan	1974	13	0.5308560	0.6180386
Afghanistan	1975	13	0.8453635	0.5308560
Afghanistan	1976	13	0.9071743	0.8453635
Afghanistan	1977	12	0.8998238	0.9071743
Afghanistan	1978	14	0.6847610	0.8998238

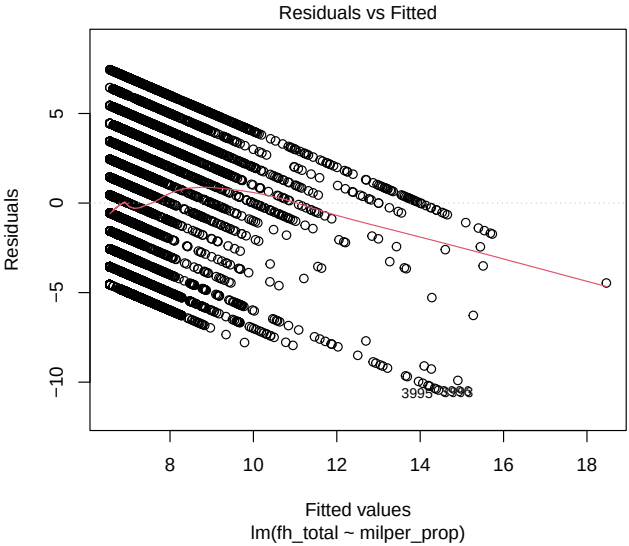
To Start

Everybody fit, evaluate and make a marginal effects plot for the first model.

1. Regress `fh_total` on `milper_prop`

	FH Total
Military Personnel (% Pop)	1.55*
	(0.06)
Constant	6.56*
	(0.06)
Observations	7,457
Adjusted R^2	0.08
Note:	*p $<$ 0.05

1. Logical?
2. Coefficients significant?
3. Adjusted R^2
4. Residuals?


Residuals vs Fitted
Residuals
5
0
−5
−10
8
10
12
14
16
18
3995
3996
Fitted values
lm(fh_total ~ milper_prop)

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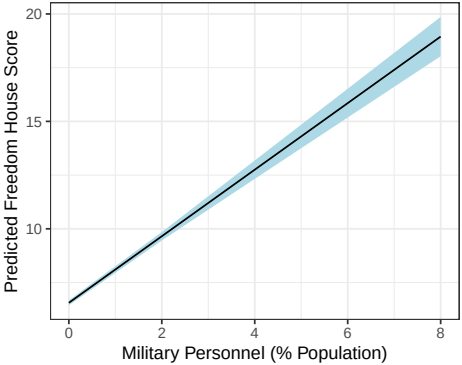

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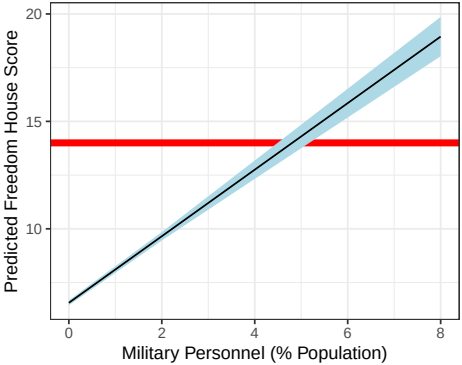

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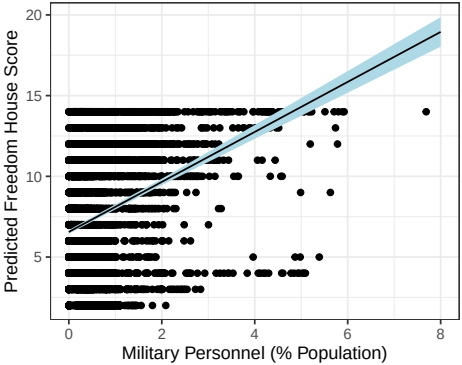

Fortunately, only 23 / 9k cases have milper_prop values above 5%

Research Project: Simple OLS Regressions

Fit, evaluate and make predictions for the following regressions:

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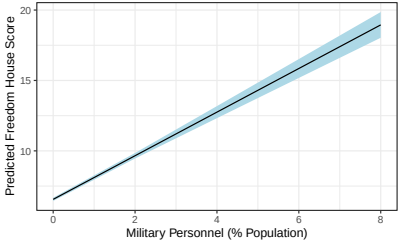
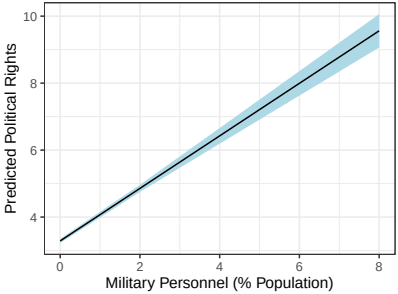
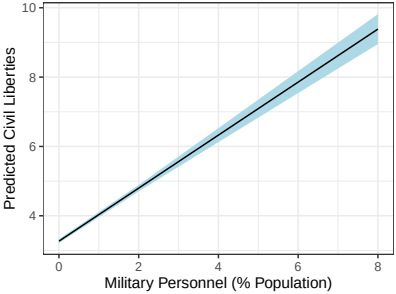
	FH Total	Political Rights	Civil Liberties
	(1)	(2)	(3)
Military Personnel (% Pop)	1.55*	0.78*	0.76*
	(0.06)	(0.03)	(0.03)
Constant	6.56*	3.29*	3.27*
	(0.06)	(0.03)	(0.03)
Observations	7,457	7,457	7,457
Adjusted R^2	0.08	0.07	0.08
F Statistic (df = 1; 7455)	614.73*	529.75*	670.48*
Note:			*p $<$ 0.05

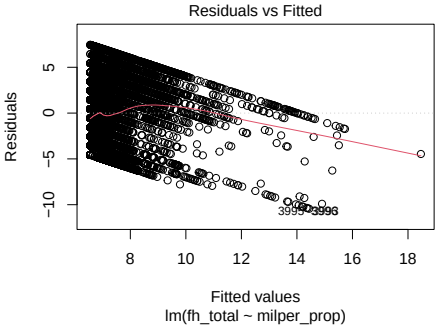
	FH Total		Political Rights		Civil Liberties	
	(1)	(2)	(3)	(4)	(5)	(6)
Military Personnel (% Pop)	1.55*		0.78*		0.76*	
	(0.06)		(0.03)		(0.03)	
Time lag = 1		1.52*		0.77*		0.75*
		(0.06)		(0.03)		(0.03)
Constant	6.56*	6.56*	3.29*	3.29*	3.27*	3.27*
	(0.06)	(0.06)	(0.03)	(0.03)	(0.03)	(0.03)
Observations	7,457	7,596	7,457	7,596	7,457	7,596
Adjusted R^2	0.08	0.07	0.07	0.06	0.08	0.08

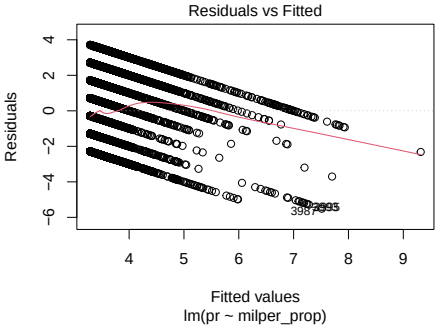
Note: *p < 0.05

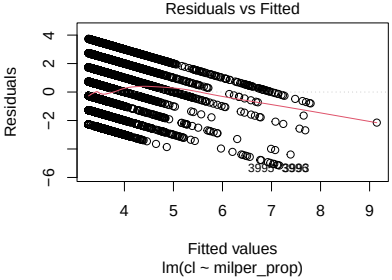

Predicted Freedom House Score
20
15
10
0
2
4
6
8
Military Personnel (% Population)


Predicted Political Rights
10
8
6
4
0
2
4
6
8
Military Personnel (% Population)


Predicted Civil Liberties
10
8
6
4
0
2
4
6
8
Military Personnel (% Population)


Residuals vs Fitted
Residuals
Fitted values
lm(fh_total ~ milper_prop)


Residuals vs Fitted
Residuals
Fitted values
lm(pr ~ milper_prop)


Residuals vs Fitted
Residuals
Fitted values
lm(cl ~ milper_prop)

For Tuesday

1. Wheelan chapter 5 “Probability”
2. Wheelan chapter 6 “Problems with Probability